Learning Degree-Generalizable Distributed Graph Coloring

Research manuscript — authors to be specified

October 2026 — preregistered experimental study

Abstract

Can a learned distributed algorithm operate at graph degrees and palette sizes absent from training? We study synchronous $(\Delta+1)$-coloring using a shared node–color network, Universal Color GNN (UCGNN). Its parameter count is independent of the palette, while its local computation and communication costs grow with it. Historical experiments with five networks trained only on degree-three, 40-node graphs recorded 8,629 successful extreme-study executions, with unequal coverage across degrees. These observations motivate a preregistered mechanism study rather than a universal-correctness claim. We isolate prefix normalization at fixed tensor shapes and initialization, compare MAX with SUM and MEAN, and implement prefix-free DeepSets and local palette attention. Across five new training seeds per primary architecture, NORMALIZED, DeepSets and local palette attention each completed all 1,520 held-out graph–tape evaluations, including degree 200 and 640-node graphs. Thus the original ordered prefixes are not necessary in this experiment. The isolated raw-endpoint control lost progress in 346 evaluations; SUM stalled in 1,196, while MEAN produced 839 improper final colorings. A finite-type reduction and interval certificates characterize the MAX COMMIT channel. A separate diffuse-component condition yields a conditional progress bound even when some colors have large probability. Neither finite success nor local safety certifies universal termination of a learned policy. A separate frozen-weight extension using only local-degree palettes recorded normalized 760/760, deepsets 760/760, transformer 760/760 proper completions. Conditional-law tests distinguish pointwise fidelity from candidate-distribution similarity.

1 Introduction

Classical distributed algorithms specify computations that apply to whole families of graphs. A learned implementation must additionally select a representation whose behavior remains meaningful when neighborhood size changes. Neural algorithmic reasoning investigates learning executable computations and their transfer across inputs [12]; the CLRS benchmark makes this challenge experimentally accessible [13]. Architectural alignment can affect extrapolation, but expressivity and training success do not by themselves establish out-of-distribution correctness [14].

We ask: *Which architectural mechanisms enable learned distributed algorithms to generalize beyond the graph degrees and palette sizes observed during training?* Our case study uses anonymous synchronous nodes, private randomness, two neighbor communications per protocol round, and permanent color decisions. We study a valid coloring with a prescribed palette, rather than optimizing the chromatic number. Local neural computations are simulated on a GPU; we do not claim a network-deployed distributed implementation.

The historical UCGNN result is striking but insufficient to identify causes: five networks trained with four colors produced proper colorings at much larger palettes, while exact candidate imitation degraded. A historical raw prefix variant concentrated proposals and stalled, but also changed the scorer dimension and initialization. Its failure cannot isolate normalization. Similarly, correct

ordinary rollouts can hide an unsafe authorization rule when candidate selection rarely exposes conflicts.

Our contribution is an auditable mechanism study: (i) tensor-, data- and tape-matched endpoint interventions, (ii) five-seed aggregation and palette controls, including successful prefix-free attention, (iii) independent graph evaluation with shared-instance statistical accounting, and (iv) separate conditional arguments for safety and progress. The outcomes support normalization and multiplicity control for this training procedure, while refuting a need for the original prefix mechanism within the tested family. We make no claim that UCGNN is inherently superior to attention, or that these architectural ingredients are necessary for every learnable coloring policy.

Related work. Message-passing GNNs and distributed local algorithms share locality constraints, although identifiers, message sizes, synchrony and stopping rules matter [4, 7, 10]. Flood and Echo Net aligns propagation with a distributed execution pattern [5]; work on asynchronous GNN inference emphasizes that synchronous performance does not imply asynchronous robustness [9]. Distributed scheduling with learned embeddings followed by a greedy solver is a different interface from learning the entire local decision policy [16]. GNN coloring work has studied decision classification and extraction of constructive solutions [3]; this does not constitute anonymous synchronous protocol execution under our observation constraints. Transformer length generalization depends on the learned computation and input representation [2, 17]. Finally, imitation learning faces policy-induced distribution shift [6]; targeted intervention can expose confounded decision rules [1]. Our counterfactual proposal supervision is specified below and should not be equated with either of these algorithms.

2 Problem and Distributed Model

Let $G = (V, E)$ be a finite simple undirected graph, $N = |V|$, with maximum degree at most a declared Δ. Nodes share the palette $\{0, \dots, K-1\}$, $K = \Delta + 1$, and maintain $q_v \in \{-1, 0, \dots, K-1\}$, where -1 is pending. No unique node identifier enters the policy. Colors have common names and an order, so color permutation invariance is not assumed. Each round draws independent $r_v^p, r_v^c \sim U[0, 1)$ even for fixed nodes. After a first neighbor communication the node selects participation B_v and a candidate C_v. A second communication permits a learned COMMIT decision J_v. Participating pending nodes with $J_v = 1$ fix C_v permanently.

The teacher's available set and decisions are

$$A_v = \{c : \forall u \in N(v), q_u \neq c\}, \qquad k_v = |A_v|, \tag{1}$$

$$B_v^* = \mathbf{1}[q_v = -1]\mathbf{1}[r_v^p < 1/2], \qquad C_v^* = A_v[\lfloor r_v^c k_v \rfloor], \tag{2}$$

where A_v is sorted. The teacher confirms only if no fixed neighbor has the candidate and no participating pending neighbor proposes it. We call this the random-trial teacher; historical code sometimes uses "Luby" as shorthand. Exact availability and conflict checks supply labels or observe completed model decisions; they never enter learned inference as a shield.

3 Supervised Learning of Distributed Coloring

Each training example is a round from a teacher trajectory, including the current partial coloring and its two private draws. The loss is

$$\mathcal{L} = \mathcal{L}_{\text{participate}} + \mathcal{L}_{\text{candidate}} + \tfrac{1}{2}\mathcal{L}_{\text{commit,own}} + \tfrac{1}{2}\mathcal{L}_{\text{commit,cf}}.$$

Participation and COMMIT use binary cross-entropy; the candidate uses cross-entropy over the public palette. During ten warmup epochs COMMIT sees teacher proposals. Thereafter it sees hard own proposals and intervened proposals with recomputed exact authorization labels. Interventions include proposing a fixed neighbor's color and equal simultaneous neighboring offers; their frozen probabilities are 0.17 and 0.04. Only proposals are intervened upon: permanent colors and the observation interface remain unchanged. Hard proposals prevent the COMMIT losses from updating candidate or participation modules. Own-only replaces the counterfactual half by own proposals, preserving the total loss weighting.

M6B uses the earlier M4B neural architecture specialized to a fixed palette. With its reference dimensions it has $385K + 7427$ parameters. Changing K changes input and output matrices, so these checkpoints are evaluated only at compatible degrees, without undocumented padding or adaptation.

4 Universal Color GNN

For every pair (v, c) define $u_v = \mathbf{1}[q_v = -1]$ and $o_{vc} = \mathbf{1}[q_v = c]$. Shared learned MLPs construct the first message and state:

$$m^1_{vc} = \phi_1([u_v, o_{vc}]), \qquad h_{vc} = \psi_1([u_v, \mathrm{MAX}_{w \in N(v)}\, m^1_{wc}]).$$

Both have 32-dimensional nonnegative outputs. Empty-neighborhood aggregation is zero. Learned availability and ordered endpoints are

$$\alpha_{vc} = \sigma(a^\top h_{vc} + b), \qquad S_{vc} = \sum_{j<c} \alpha_{vj}, \qquad D_v = \max(\sum_j \alpha_{vj}, 0.5), \tag{3}$$

$$L_{vc} = S_{vc}/D_v, \qquad U_{vc} = (S_{vc} + \alpha_{vc})/D_v, \tag{4}$$

$$s_{vc} = F([h_{vc}, \alpha_{vc}, L_{vc}, U_{vc}, r^c_v]). \tag{5}$$

The candidate is the maximizing palette slot; there is no exact mask of occupied colors. The availability scores need not be calibrated probabilities or an integer count. Prefix accumulation and normalization are incorporated operations, while availability, scoring and authorization are learned.

For the second communication let

$$f_{vc} = [u_v, o_{vc}, B_v, \mathbf{1}[C_v = c]], \qquad z_{vc} = \psi_2([f_{vc}, \mathrm{MAX}_{w \in N(v)}\, \phi_2(f_{wc})]).$$

The COMMIT head reads only z_{vC_v} and the own participation/pending/offer bits. The original model has 16,035 trainable parameters independent of K. The implemented messages contain $32K$ floats per directed edge per stage; constant parameter count is not constant communication or memory cost.

5 Prospective Experimental Protocol

The protocol was locally committed as `023deaf` before new training or confirmatory test generation. Primary treatments are NORMALIZED, raw endpoints, SUM, MEAN, DeepSets and LocalPaletteTransformer. The raw-endpoint control changes only (L, U) to $(S, S + \alpha)$ in the same four extra scorer inputs. It shares every initial tensor, example, minibatch order, counterfactual tape, optimizer and update count with NORMALIZED within a training seed. SUM and MEAN replace the

Model	Networks	Executions	Successes	Graphs	Unsafe
A1	1	400	357	400	0
C1	1	400	400	400	0
E1	5	8629	8629	1765	0
E2	5	2000	2000	400	0
E3	5	2000	2000	400	0

Table 1: Historical extreme-study counts. Coverage differs between experiments; shared graphs do not become independent by evaluating more networks.

neighbor reduction in both stages, without extra scaling for SUM; MEAN uses actual in-degree and zero for an isolated node.

Training uses 800 regular graphs with $N = 40$, $\Delta = 3$, three tapes, Adam learning rate 10^{-3}, batch size 128 and 120 epochs. New training seeds are 41000–41004. Seed 41000 is the full-budget pilot and counts as the first replicate. Promotion requires final training-degree validation accuracies of at least 0.98 participation, 0.90 candidate and 0.95 own-COMMIT, with finite training throughout. Failed pilots are retained without repeated tuning. Promotion decisions precede the confirmatory test. Epoch 120 is primary; validation-best is a separately labeled selection policy.

The independent namespace 202610210211 fixes 304 graph–tape cases per model: degrees 3,5,10,15,20,50,100,200, regular/irregular/heterogeneous families, $N = 160$ (400 at degree 200), size checks at $N = 640$, cliques and circulant bipartite stress graphs. Each random core cell contains eight graphs; size cells contain four. A clique's three tapes are not three independent graphs. Every paired model uses the same instances and tapes, with a 256-round cap. Failures do not stop other evaluations or truncate a rollout at its first unsafe action. Technical OOM and timeout remain distinct from wrong coloring.

We report fixed-network Clopper–Pearson intervals per random-graph stratum, all seed rates, and paired hierarchical bootstrap differences across training seeds and graph identities. Correlated executions on shared graphs are not pooled as independent Bernoulli trials. Computational limits are an 8 GiB RTX 4060, one GPU worker, a 4 GiB allocator cap, and 36 accumulated active GPU-process hours. Source hashes, optimizer/RNG checkpoints and atomic per-case records support recovery. The full preregistration specifies the remaining metrics and resource rules. All 157 scheduled tasks completed in 6.172 accumulated GPU-process hours, well below the 36-hour ceiling. The release contains 32 new training runs, 85 checkpoint views, 18,363 graph evaluations and 3,520 controlled progress rollouts. There were no evaluation OOMs, timeouts or unexecuted planned cells. The 32 runs comprise 30 primary networks and two one-seed auxiliary pilots; validation-selected views do not count as additional independent networks. At eight random graphs per core stratum, an all-success fixed-network cell still has a Clopper–Pearson lower bound of only about 0.631. Empirical bootstrap intervals can be degenerate when all observed differences agree; this does not establish population certainty or equivalence.

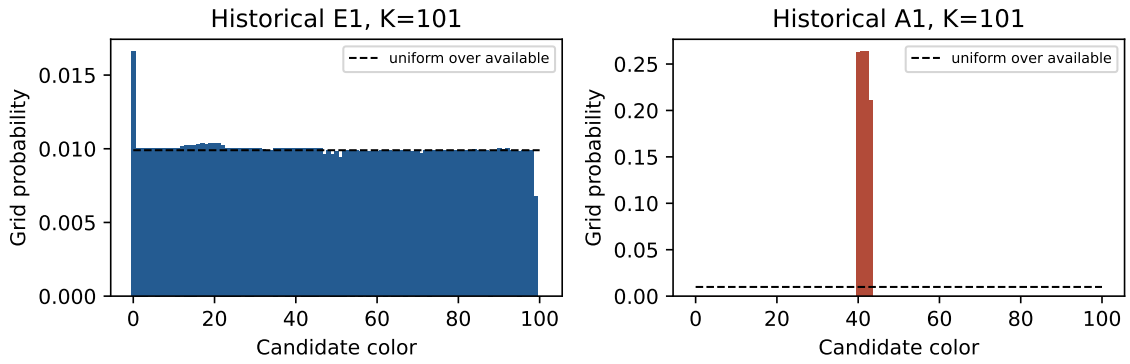


Figure 1: Historical candidate distributions, generated from saved per-color CSVs. These descriptive observations do not isolate the effect of normalization.

6 Degree and Size Generalization: Historical Evidence

Five historical E1 networks trained solely at degree three recorded 8,629/8,629 successes in the extreme study, on 1,765 distinct graphs with unequal coverage across degree, size and seed. The maximum degree 500 was exploratory with sparse coverage, not a balanced five-network test at that degree. Historical E2 and E3 each had five independent training networks; A1 and C1 each had one. These already observed cohorts are motivation and historical evidence, not untouched architecture-selection tests.

E1-0's exact candidate fidelity can collapse while valid coloring remains successful. In a historical all-free $K = 201$ controlled sweep it agreed with the teacher at only 5/1,001 uniform-grid draws while using 199 colors. Thus candidate identity and diversity are different endpoints. Historical A1's all-free $K = 101$ law placed mass on only four colors. Reconstructed failures showed extremely slow progress with no unsafe commits, localizing a functional failure without isolating which architectural change caused it.

Independent new cohort. Each of five new NORMALIZED, DeepSets and Transformer networks completed 304/304 evaluations with no exact-rule-violating COMMIT, across all three random families, size checks and structured graphs. Historical E1 also completed 1,520/1,520 new-cohort executions. The 15 M6B specialists, five at each compatible degree 3, 5 and 10, completed 555/555 compatible tests; their fixed-palette heads were not evaluated at unseen degrees. They were trained at their evaluation degree, unlike the universal models' degree-transfer setting.

7 Architectural Ablations

All five normalization pairs passed the complete tensor, data, minibatch, counterfactual-tape and update-count audit. Raw endpoints completed 1,174/1,520 executions, with five-network rates ranging from 0.766 to 0.780, and left nodes pending in 346 capped runs. No unsafe COMMIT was observed. At all-free $K = 201$, its effective support was only 4.00–4.05 colors, versus 193.1–199.5 for NORMALIZED. This isolated intervention supports a beneficial normalization effect under the fixed procedure; it does not establish that all raw parameterizations must fail.

SUM completed 324/1,520 tests, with no completion at any tested degree at least 10. Its 1,196 failures reached the round cap without an unsafe COMMIT. The multiplicity probe showed

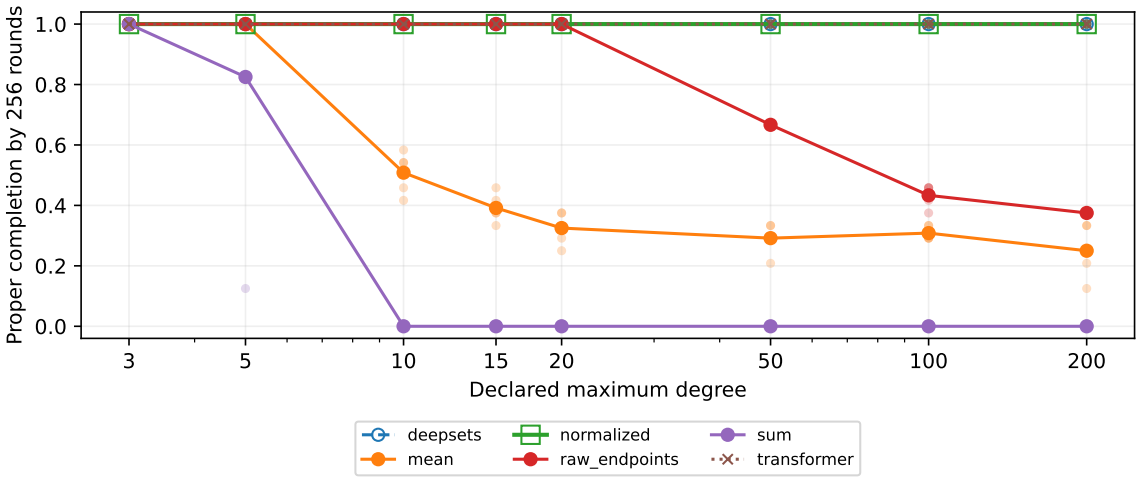


Figure 2: Core-tier proper completion by 256 rounds. Lines average five independently trained networks; individual seed rates are overlaid. All architectures use the same graphs and tapes.

Model	Networks	Proper / executions	Seed rate range	Unfinished	Unsafe COMMIT
normalized	5	1520/1520	1.000–1.000	0	0
raw_endpoints	5	1174/1520	0.766–0.780	346	0
sum	5	324/1520	0.145–0.230	1196	0
mean	5	681/1520	0.411–0.487	0	57340
deepsets	5	1520/1520	1.000–1.000	0	0
transformer	5	1520/1520	1.000–1.000	0	0

Table 2: Primary epoch-120 algorithmic outcomes across all strata. Totals are descriptive: networks share 288 graph identities and 304 graph–tape cases. Technical failures are reported separately. Unsafe denotes violation of the exact rejection rule; final improper coloring is assessed separately.

rejection of conflict-free proposals after duplicating an ordinary neighbor type eight times, for all five networks. MEAN completed 681/1,520; every node eventually fixed a color, but 839 final states were improper. Its COMMIT heads missed single conflict messages diluted by eight ordinary messages in all five networks. Both aggregations learned the degree-three teacher well. These observations separate over-rejection from unsafe acceptance and support the importance of multiplicity control in these models. Since both communication stages were changed together, the graph ablation alone does not identify which stage caused every error.

The separately selected validation-best checkpoints retain the same qualitative pattern on 960 core executions per primary treatment: NORMALIZED, DeepSets and Transformer each 960/960, raw 799/960, SUM 255/960 and MEAN 494/960. Selection did not use held-out degrees.

8 Transformer and Other Baselines

LocalPaletteTransformer keeps shared local graph communication and the COMMIT pathway, replacing the prefix scorer by a 16-dimensional, two-head within-node self-attention block with feedforward width 32 and zero dropout [11]. It receives functional positions $c/\max(K-1,1)$ and $1/K$, not learned absolute position embeddings or original prefixes. Its parameter count is 16,995.

Model	Networks	3	5	10	15	20	50	100	200
cyclic_teacher	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
deepsets	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
mean	5	1.0000	1.0000	0.5083	0.3917	0.3250	0.2917	0.3083	0.2500
normalized	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
own_only	1	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000
raw_endpoints	5	1.0000	1.0000	1.0000	1.0000	1.0000	0.6667	0.4333	0.3750
sum	5	1.0000	0.8250	0.0000	0.0000	0.0000	0.0000	0.0000	0.0000
transformer	5	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000	1.0000

Table 3: Core-tier proper completion, means across independent training networks (columns are degrees). Every network uses shared graphs. Seed ranges, paired intervals and coverage are in the accompanying CSVs.

Model	Params	Dense GFLOPs	Allocated MiB	Round ms
deepsets	17907	2.4126	269.5396	46.0643
mean	16035	2.1168	471.8867	76.5871
raw_endpoints	16035	2.1168	269.1333	45.6381
transformer	16995	3.2797	345.3462	49.9987
sum	16035	2.1168	471.8867	76.0992
normalized	16035	2.1168	269.1333	45.3488

Table 4: One-round resource probe at N=400, degree 200, K=201 (one prespecified seed per model). FLOPs are approximate; neighbor reductions, nonlinearities and overhead are itemized separately in CSV. CUDA allocated memory excludes driver/reserved memory.

The DeepSets control uses a shared 16-dimensional mean-pooled palette context [15], functional positions and 17,907 parameters. Attention can have quadratic palette cost; a technical resource limit must not be called a coloring failure. All six seed-41000 pilots reached epoch 120 and passed the prespecified local learning gate. Candidate validation fidelity was 0.9971 for NORMALIZED, 0.9850 for raw endpoints, 0.9983 for SUM and MEAN, 0.9948 for DeepSets and 0.9957 for the Transformer; own-proposal COMMIT fidelity was 1.0 for each. These are in-degree validation measurements, not degree-transfer results. In a separate inference-only resource check, all six models fit at $N = 640$, $\Delta = 200$ and $K = 201$. The Transformer used approximately 532 MiB of peak allocated CUDA memory for one round; this establishes technical feasibility for this configuration only. Both prefix-free controls also completed all 205 degree-200 evaluations across five seeds and all strata. This directly answers the palette transfer question affirmatively for $K = 11, 21, 51, 101, 201$ on the stated cohort. However, attention was less uniform and slower on large cliques: at all-free $K = 201$ its effective support ranged from 57.3 to 112.9 colors, with largest grid masses 0.126–0.243. On the degree-200 clique its mean termination time was 30 rounds, versus 14.33 for NORMALIZED and 21.47 for DeepSets (five networks, three shared tapes each; descriptive means). DeepSets' effective support was 128.6–174.2. Exact teacher-candidate fidelity was low for these successful policies; fidelity is not a coloring guarantee. Small numbers of forbidden proposals occurred for two Transformer seeds and one DeepSets seed, but their certified COMMIT channels rejected them.

These are architectural controls, not a claim that the only possible prefix-free model has been

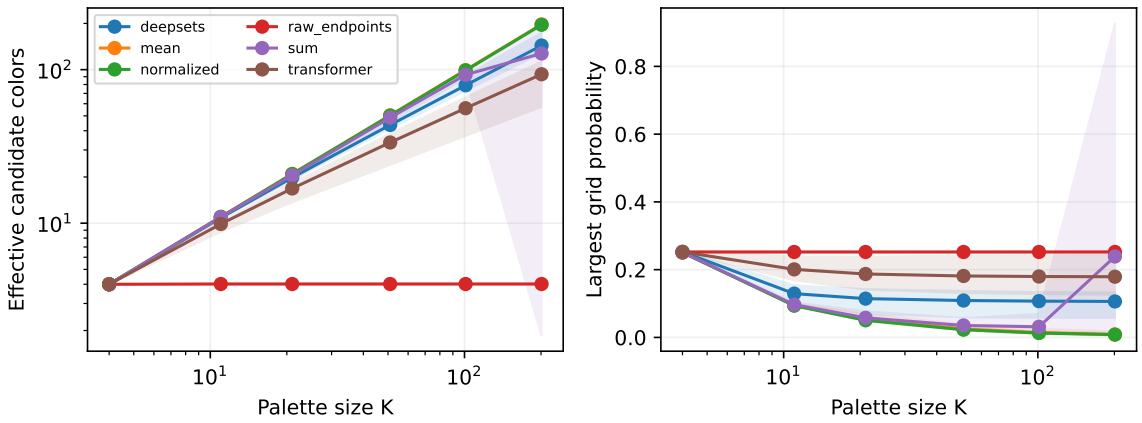


Figure 3: All-free candidate laws from 2,049 midpoint draws. Lines are five-network means; shaded bands show seed ranges, not confidence intervals. The largest atom alone does not capture the diffuse tail.

tested. Without positions or another symmetry breaker, identical available slots receive identical scores under an invariant pooled context, preventing general ordered-teacher imitation.

9 Understanding Learned Policies

9.1 Selection and held-out prediction

The prospective diagnostic freezes a monotone rank calibration of historical E1-0 on all-free $K = 101$ before evaluating $K = 17, 63, 251, 997$ and low, high, scattered, alternating and extreme availability patterns. It reports rank errors, forbidden proposals, total variation from uniform, entropy, effective colors, maximum mass and pair collisions. Boundary-neighbor float32 probes are separate from the midpoint grid used for distribution estimates. The CPU float32 diagnostic, prospectively authorized in addendum `8c69168`, evaluated 120 configured states with 2,049 midpoint draws each. Equivalent all-free configurations occur under multiple pattern names; these are not 120 independent random states. No forbidden candidate was observed on these grids. The frozen empirical predictor's mean per-state rank error was 0.569; 10 configurations at $K = 997$ exceeded two, with a worst per-state mean of 4.761. Thus its approximation deteriorates at the largest palette despite a small pooled mean. We retain these negative cases.

Two secondary analytic predictors were also frozen before held-out inspection. The context-conditioned predictor uses a finite catalog of neighbor-type contexts calibrated at $K = 101$. It achieved mean rank error 0.075 and worst per-state mean 0.236, with individual grid rank error at most one. This is finite empirical evidence for a context-dependent deformed quantile, not an exact algorithmic identity or a guarantee over all availability patterns. Specifically, fix the learned free-color context h_s, α_s and set both endpoints to a continuous coordinate t. The one-hidden-layer scorer becomes

$$f_s(t, r) = b_0 + \sum_j v_j \,\mathrm{ReLU}(B_{sj} + A_j t + R_j r).$$

Its smallest continuous maximizer $g_s(r)$ is found among $0, 1$ and the in-range breakpoints $-(B_{sj} + R_j r)/A_j$ for $A_j \neq 0$. Predicting rank $\lfloor k g_s(r) \rfloor$, clipped to the available range, removes the empirical calibration grid's quantization. The four contexts encode presence of pending and fixed neighbors.

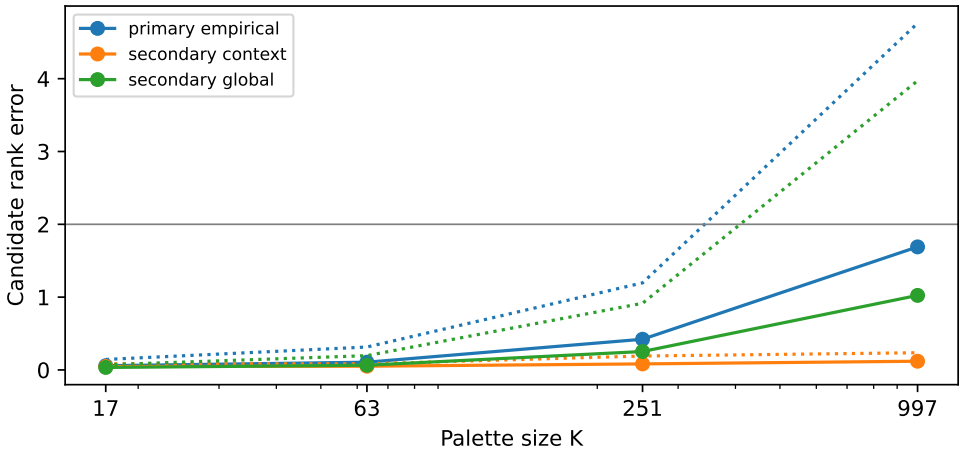


Figure 4: Frozen E1-0 candidate predictions from saved CSVs. Solid lines show mean per-state rank error; dotted lines show the worst per-state mean. Primary and secondary predictors remain separately identified.

This is a weight-derived predictor, evaluated numerically in float64; replacing finite-width intervals and neglecting forbidden-color availability mass are approximations, not a convergence theorem. The single global-context predictor remains separately reported.

9.2 A sufficient condition for progress

Condition on a proper state. Let U_v be the pending neighbors. With private independent participation/candidate draws within and across nodes, and exact conflict rejection,

$$\Pr(v \text{ confirms} \mid \text{state}) = p_v \sum_{c \in A_v} q_v(c) \prod_{u \in U_v} (1 - p_u q_u(c)). \tag{6}$$

Confirmation events of neighboring nodes need not be independent.

Proposition 1. *Suppose* $0 < p_{\min} \le p_u \le p_{\max} < 1$, *exact conflict rejection and the independence assumptions of (6) hold. If* $\operatorname{supp}(q_v) \subseteq A_v$ *and* $\max_c q_v(c) \le \beta/k_v$, *then*

$$\Pr(v \textit{ confirms} \mid \textit{state}) \ge p_v(1 - p_{\max})^{\beta |U_v|/k_v} \ge p_{\min}(1 - p_{\max})^{\beta}.$$

Proof. Properness and $K \ge \Delta + 1$ imply $k_v \ge |U_v| + 1$. For $0 \le x \le p_{\max}$, concavity gives $\log(1 - x) \ge x \log(1 - p_{\max})/p_{\max}$. Set $\Lambda_c = \sum_{u \in U_v} p_u q_u(c)$ and $a = \log(1 - p_{\max})/p_{\max} < 0$. The product in (6) is at least $\exp(a\Lambda_c)$. Jensen's inequality gives $\sum_c q_v(c) e^{a\Lambda_c} \ge e^{a \sum_c q_v(c)\Lambda_c}$. Moreover, $\sum_c q_v(c)\Lambda_c \le (\beta/k_v) \sum_u p_u \le \beta p_{\max} |U_v|/k_v$. Since $a < 0$, substitution proves the first bound; $|U_v|/k_v \le 1$ proves the second. □

If these hypotheses hold uniformly at every reachable state for every pending node, $\rho = p_{\min}(1 - p_{\max})^{\beta}$ gives $\Pr(T > t) \le \min\{1, N(1-\rho)^t\}$ by conditional iteration and union bound. We have not established the uniform candidate hypotheses for any learned checkpoint. In contrast, proposing one common deterministic color on an all-free K-clique with independent participation $1/2$ is safe but has mean time $2^K/K$ until the first confirmation. Uniform *marginals* with a shared candidate draw have the same failure of progress, showing why conditional independence matters.

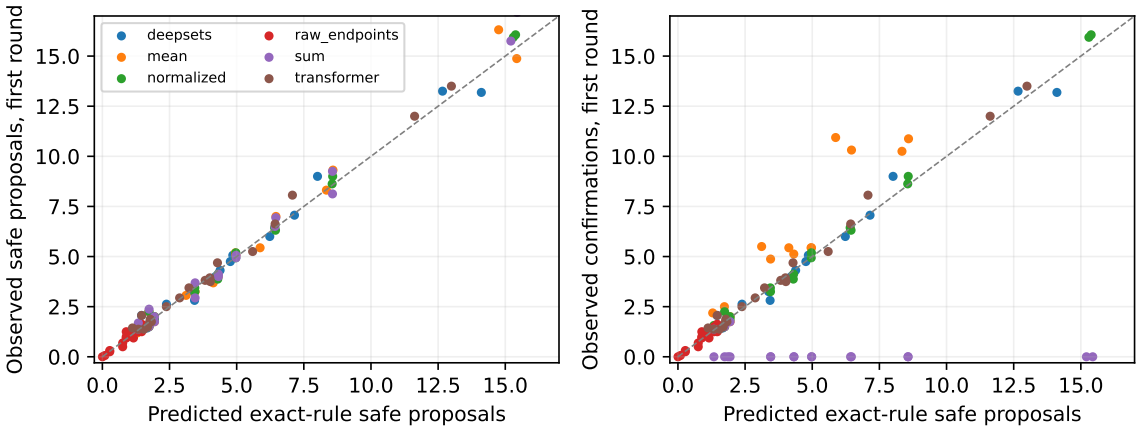


Figure 5: Controlled first-round predictions, using midpoint-grid estimates of p, q, versus 16-tape averages. Safe proposals and actual confirmations coincide only for an exact COMMIT channel. States within a network are descriptive probes, not independent trained networks.

Corollary 1 (A diffuse allowed component)**.** *Retain properness, exact rejection, independent private draws and the participation bounds. If $S_v \subseteq A_v$ has mass $\eta_v = \sum_{c \in S_v} q_v(c) > 0$ and $q_v(c) \le \beta/k_v$ on S_v, then*

$$\Pr(v \text{ confirms} \mid \text{state}) \ge p_v \eta_v (1 - p_{\max})^{\beta |U_v|/(\eta_v k_v)}.$$

If $\eta_v \ge \eta > 0$ uniformly, this is at least $p_{\min}\eta(1 - p_{\max})^{\beta/\eta}$.

Proof. Drop terms outside S_v in (6) and normalize the retained weights to $q_v(c)/\eta_v$. The same log-chord and Jensen inequalities give $p_v \eta_v \exp(a \sum_{c \in S_v} q_v(c)\Lambda_c/\eta_v)$. Its averaged load is at most $\beta p_{\max}|U_v|/(\eta_v k_v)$, proving the first inequality. Use $|U_v|/k_v \le 1$ and monotonicity of $x(1-p_{\max})^{\beta/x}$ for $x > 0$ for the second. □

This post-result mathematical extension permits a heavy atom outside S_v, and even forbidden mass rejected by COMMIT. It helps distinguish a concentrated policy with a diffuse tail from one supported on four colors. It remains a sufficient condition; grid measurements do not establish it uniformly for a learned network.

Controlled states. Eleven prespecified representative checkpoints underwent 20 proper partial states with 16 common tapes each. NORMALIZED, DeepSets and Transformer each completed 320/320, versus raw 224/320 and SUM 64/320. MEAN completed 152/320 properly; historical A1 completed all 320 but needed up to 159 rounds. First-round safe proposal counts tracked the independent-choice expression, while SUM could reject abundant safe proposals and MEAN could confirm unsafe ones. These distinguish candidate quality from learned authorization.

9.3 Finite-type safety of COMMIT

Proposition 2. *For a legal uncolored participating center with candidate c, nonnegative shared messages and MAX aggregation, the UCGNN COMMIT logit depends only on the set of six legal neighbor types in slot c. Correct strict signs on all 64 type sets imply exact local conflict rejection for arbitrary degree and palette in real arithmetic.*

treatment	Networks	Applicable	Certified	Unsafe_masks	Margin_min
A1	1	1	1	0	11.7952
C1	1	1	0	16	-15.3640
E1	5	5	5	0	9.7060
cyclic_teacher	1	1	1	0	10.7750
deepsets	5	5	5	0	8.6519
mean	5	0	0	0	NaN
normalized	5	5	5	0	8.6298
own_only	1	1	0	16	-19.3702
raw_endpoints	5	5	5	0	8.6507
sum	5	0	0	0	NaN
transformer	5	5	5	0	8.6784

Table 5: Finite-type real-arithmetic certificates, with explicit arithmetic assumptions. Unsafe mask totals across networks are descriptive. SUM/MEAN are outside this certificate's scope.

Proof. The legal types $[u, o, B, \text{offer}]$ are

$$(0,0,0,0), (0,1,0,0), (1,0,0,0), (1,0,0,1), (1,0,1,0), (1,0,1,1).$$

Colored neighbors cannot participate or offer; pending neighbors do not own a fixed color. MAX is idempotent, so multiplicities disappear. The own candidate gather reads one slot, whose own features are $(1,0,1,1)$; other colors cannot alter that logit. Fixed and simultaneous conflicts correspond precisely to the second and sixth types. The 64 strict sign conditions therefore suffice. Other own states are prevented from execution by explicit protocol gating. All subsets are realizable by proper star states with a palette of at least seven colors. □

Induction then preserves properness, without proving progress or termination. All five historical E1 checkpoints now have interval-certified strict signs on all 64 sets, both for real weights and under the stated float32 affine-error model. Their real-arithmetic minimum signed margins range from 9.706 to 13.547. Historical C1 and the new own-only seed-41000 checkpoint each have 16 certified unsafe type sets. The new NORMALIZED, raw-endpoint, DeepSets and Transformer seed-41000 COMMIT channels also satisfy the certificate; this does not certify their candidate selection. SUM and MEAN lack this 64-set reduction, which is an applicability limitation rather than evidence of unsafe decisions. Complete arithmetic assumptions, per-mask intervals and checkpoint hashes accompany the certificate CSV and JSON records. The completed audit covers 39 checkpoints: 27 satisfy the real and conditional float32 exact-rule certificate, two own-only checkpoints fail, and ten SUM/MEAN checkpoints are outside its scope. Actual star-forward decisions agree with the type reduction in all 2,496 tests. All 1,856 applicable float32 logits lie within their error enclosures. The 27 certified models also preserve signs under the sampled $\pm 10^{-6}$ pooled-message perturbations in float32 and float64; conservative real-arithmetic input radii are positive. This certifies a local rejection rule under the stated assumptions, without a global progress guarantee.

The new own-only pilot and historical C1 each completed 304/304 ordinary graph tests and 320/320 controlled progress rollouts despite their 16 unsafe type sets. Their proposals did not expose these errors. Counterfactual supervision therefore improves the learned off-trajectory COMMIT rule in these comparisons; ordinary successful execution alone cannot certify it.

A structural limitation of MEAN. Fix the active center's own features. Let m_o, m_f be degree-independent messages for an ordinary neighbor and a fixed-color conflict, respectively, and F the continuous post-aggregation COMMIT logit with no separate degree or palette input. This is the implemented per-slot interface.

Proposition 3. *If $F(m_o) = \gamma > 0$, MEAN accepts a conflicting legal star at some finite degree. Thus it cannot reproduce exact conflict rejection at all degrees while strictly accepting that conflict-free state.*

Proof. With d ordinary leaves and one conflicting fixed leaf, the received mean is $x_d = (dm_o + m_f)/(d+1) \to m_o$. The center is pending and the fixed leaves are nonadjacent, so the partial state is proper with $K = d + 2$. Continuity implies $F(x_d) \to \gamma > 0$, hence eventual unsafe acceptance. For an L-Lipschitz classifier, it suffices that $d + 1 > L\|m_f - m_o\|_\infty/\gamma$. □

This is a local expressivity result under the stated interface, not a claim that the model reaches every such state. SUM has a contrasting construction: on legal types, $b(u, o, p, a) = o + \mathrm{ReLU}(p + a - 1)$ is exactly the conflict bit, so $1/2 - \sum_w b_w$ implements exact rejection. Its observed failure is therefore not an impossibility of representing this rule.

10 Removing Global Degree Knowledge

The preceding experiments prescribe a common palette using the graph's maximum degree. That information need not be locally available. We now replace it by the exact number of neighbors, using the individual list $L_v = \{0, \ldots, d(v)\}$. A node may count its neighbors; no global maximum is supplied as a learned feature. Common color names remain necessary: different nodes may have different lists but agree about equality of overlapping colors.

Exact protocol. Let t_v count pending neighbors of a pending node in a proper partial coloring, and let A_v be its list after removing fixed-neighbor colors. There are at most $d(v) - t_v$ excluded colors, so

$$|A_v| \geq d(v) + 1 - (d(v) - t_v) = t_v + 1. \tag{7}$$

A fixed neighbor whose color is outside L_v excludes nothing. Consider the exact teacher with independent private participation of probability $1/2$, uniform proposals from A_v, and exact rejection of fixed or simultaneous conflicts. List membership and properness are invariant: old edges remain proper, new–old conflicts are explicitly rejected, and adjacent equal simultaneous offers are both rejected. Fixed colors never change.

Proposition 4 (Exact local-list progress)**.** *Under these hypotheses every pending vertex confirms in the next round with conditional probability at least $1/4$. For a graph on n vertices and any proper partial starting coloring, $\Pr(T_{\text{final}} > t) \leq \min\{1, n(3/4)^t\}$.*

Proof. Writing $k_v = |A_v|$ and $q_u(c) = 0$ outside a neighbor's available list, private independence gives

$$P_v = \frac{1}{2k_v} \sum_{c \in A_v} \prod_{u \in N(v) \text{ pending}} (1 - \tfrac{1}{2} q_u(c)).$$

The inequality $\prod_i (1 - x_i) \geq 1 - \sum_i x_i$ yields $P_v \geq 1/2 - (4k_v)^{-1} \sum_u \sum_{c \in A_v} q_u(c) \geq 1/2 - t_v/(4k_v) \geq 1/4$ by (7). Each pending node's conditional survival probability is thus at most $3/4$ per round. Iterated conditioning and then a union bound give the tail, without assuming independent survival events across vertices. □

These are guarantees of the exact teacher. They do not follow for neural availability or neural COMMIT merely because sampled rollouts are proper.

Frozen architecture extension. All thirty epoch-120 checkpoints are reused without retraining. The receiver evaluates its valid colors using the same shared message, state and scorer modules. A sender's pending and participation bits remain defined at a receiver color outside the sender's list, while its own-color and offer bits are zero there. Deleting that message would change the learned protocol. A fixed sender's color outside the receiver's list is represented as a fixed node with own-bit zero at every queried receiver color.

The implementation groups receivers by their own K_v. Palette reductions and attention operate at exactly that length; padding is only output storage. The second communication is evaluated at the receiver's candidate, the only slot read by the historical COMMIT head. These are algebraic restrictions of the original modules, not additional masks of exact availability or a safety shield. A finite-alphabet reduction accelerates MAX first messages without changing the message function. Floating-point action equivalence, padding, node permutation, disjoint batch composition and isolated nodes are tested separately. Real-arithmetic equivalence does not imply bitwise equivalence between differently shaped GPU kernels.

The six legal selected-color neighbor types of the MAX COMMIT channel remain the same for heterogeneous lists: excluding a color from a neighbor's list can only force its own and offer bits to zero. No K or position enters COMMIT. Consequently a verified negative logit for every conflicting one of the 64 type sets transfers to arbitrary local lists in real arithmetic. We matched the twenty primary MAX checkpoint hashes to their previous interval certificates and additionally tested 1,280 legal local-list witnesses, placing the center's candidate at its last local color. Float32 transfer retains the affine-error and no-TF32/overflow assumptions of those certificates. This is a safety argument, with no neural termination conclusion. 5/5 frozen gates have a certified error in the degree-8 all-pending-other local-star intervention. 5/5 frozen gates have a certified error in the degree-8 one-fixed-same local-star intervention. 5/5 frozen gates have a certified error in the degree-16 one-pending-same local-star intervention. These are exploratory counterexamples selected from the recorded degree-probe grid, with unchanged checkpoints. They establish errors of these learned gates in real arithmetic; they do not establish an expressivity impossibility for SUM.

The prefix-free baselines contain explicit functional position $c/\max(K_v - 1, 1)$ and inverse size $1/K_v$. These are based on a node's own list. They distinguish free colors even when all neighborhood states h_{vc} are identical. Thus their successful historical extrapolation does not show that color order was absent or that the architectures learned a fully permutation-equivariant palette policy.

Prospective comparison. The local-palette protocol was committed before confirmatory evaluation, with 30 checkpoint hashes, independent graph seeds and separate pilot seeds. It prescribes 76 graph specifications, two tapes and four regimes: GLOBAL ($K_v = \Delta+1$), LOCAL ($K_v = d(v)+1$), and common oversized lists with $K_v = \lceil a\Delta \rceil + 1$ for $a \in \{1.5, 2\}$. The eight families include regular controls, irregular and preferential-attachment graphs, stars, bipartite graphs, hubs, dense graphs and hub–low-degree mixtures, with target degrees 3, 10, 50 and 200 and additional 640-node graphs. Actual degrees are recorded. Deterministic structured graphs are descriptive controls, not independent random-graph draws. Every network/regime uses the same graph and random tape. A 256-round cutoff is an algorithmic progress limit, whereas OOM and wall-clock limits are technical failures. Unsafe authorizations, final edge conflicts and forbidden proposals are separate outcomes.

deepsets completed 760/760 local-list evaluations, with 0 exact-rule-violating authorizations. mean completed 431/760 local-list evaluations, with 18,727 exact-rule-violating authorizations.

normalized completed 760/760 local-list evaluations, with 0 exact-rule-violating authorizations. raw_endpoints completed 637/760 local-list evaluations, with 0 exact-rule-violating authorizations. sum completed 160/760 local-list evaluations, with 0 exact-rule-violating authorizations. transformer completed 760/760 local-list evaluations, with 0 exact-rule-violating authorizations. These are finite-cohort outcomes; shared graphs and five networks do not create independent graph samples. 5 specifications (4 distinct structures) are previously examined controls; 45 fresh random specifications are reported separately.

Treatment	Regime	Success / cases	Mean rounds
deepsets	GLOBAL	760/760	9.02
deepsets	LOCAL	760/760	9.17
deepsets	OVERSIZED-1.5	760/760	8.62
deepsets	OVERSIZED-2	760/760	8.47
mean	GLOBAL	431/760	8.31
mean	LOCAL	431/760	8.43
mean	OVERSIZED-1.5	453/760	8.27
mean	OVERSIZED-2	463/760	8.10
normalized	GLOBAL	760/760	8.86
normalized	LOCAL	760/760	9.03
normalized	OVERSIZED-1.5	760/760	8.60
normalized	OVERSIZED-2	760/760	8.35
raw_endpoints	GLOBAL	637/760	54.18
raw_endpoints	LOCAL	637/760	54.21
raw_endpoints	OVERSIZED-1.5	637/760	54.18
raw_endpoints	OVERSIZED-2	637/760	54.18
sum	GLOBAL	160/760	203.62
sum	LOCAL	160/760	203.58
sum	OVERSIZED-1.5	160/760	203.56
sum	OVERSIZED-2	160/760	203.43
teacher	GLOBAL	152/152	8.97
teacher	LOCAL	152/152	8.97
teacher	OVERSIZED-1.5	152/152	8.56
teacher	OVERSIZED-2	152/152	8.33
transformer	GLOBAL	760/760	9.82
transformer	LOCAL	760/760	9.78
transformer	OVERSIZED-1.5	760/760	9.26
transformer	OVERSIZED-2	760/760	9.12

Table 6: Frozen-checkpoint evaluation by palette regime. Counts include all five network seeds on shared graph–tape cases; they are not independent graph replicates. Rounds include the 256-round cutoff and improper completions; MEAN's termination time is not a speed comparison for successful proper coloring.

11 Statistical Distinguishability of Learned Coloring Policies

Successful coloring does not require exact teacher imitation. To characterize what was learned, fix the *same* local state S before comparing policies. The observer knows S, the list and the available set, observes IID candidate colors, and does not observe random draws, logits or internal activations. This differs from identifying models from their autonomously visited states.

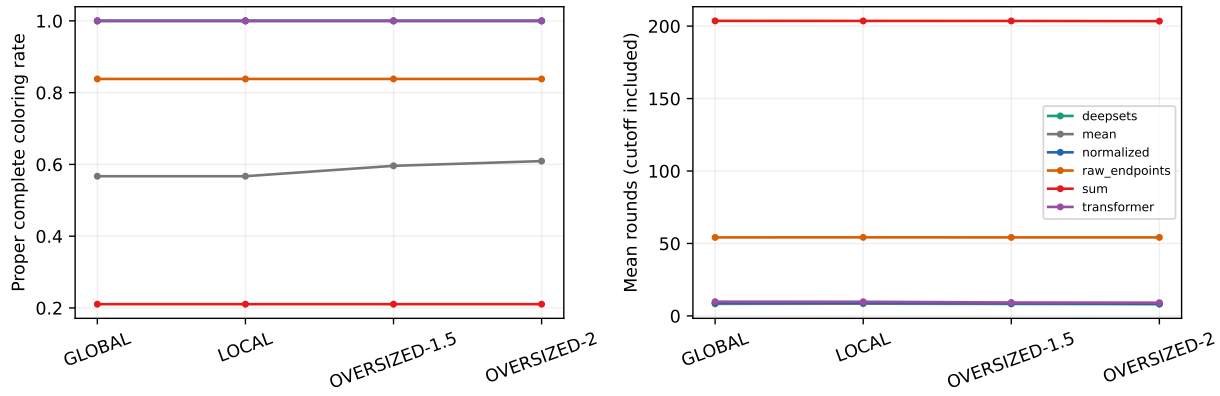


Figure 6: Success and capped rounds for the new palette regimes. Comparisons are paired by graph and private random tape.

Three distinct comparisons. Shared-randomness fidelity is $F_M(S) = \Pr[f_M(S, r_c) = f_L(S, r_c)]$. Candidate-law comparison instead concerns $q_M(c \mid S) = \Pr[f_M(S, r_c) = c]$ and the uniform teacher law on A_v. For finite-sample identification, the hypotheses are q_L^m versus q_M^m. Participation, candidates conditional on participation, and final confirmed colors must not be pooled. In these actual implementations participation sees only (r_p, u_v) and the candidate pathway sees r_c and deterministic state, so private independent draws imply conditional independence. COMMIT and confirmation additionally depend on neighboring offers.

Proposition 5 (Fidelity need not determine a candidate law)**.** *For any state with $k \geq 2$ available colors there exist two candidate maps with zero shared-r fidelity and identical uniform candidate laws.*

Proof. For sorted colors $a_0, \ldots, a_{k-1}$ take $f_L(r) = a_{\lfloor kr \rfloor}$ and $f_M(r) = a_{(\lfloor kr \rfloor + 1) \bmod k}$. They disagree everywhere on $[0, 1)$, and each color has a preimage of length $1/k$ for either map. □

Consequently no candidate-only classifier can distinguish these two laws at this state, regardless of sample size. Their equality at one state does not establish equality over every reachable state.

Transition-law equivalence. If policies share the conditional *joint* law of all relevant actions at every state and use the same update function, their next-state laws are the same pushforward of that action law. With identical initialization, induction then gives identical trajectory laws. Equality of candidate marginals alone does not suffice: two adjacent always-participating nodes with identical shared uniform candidates always collide, whereas independent uniform binary candidates collide with probability 1/2. Coupling participation to the candidate can likewise change transition laws without changing either marginal.

Distances and optimal classification. For known laws P, Q and equal priors, optimal balanced accuracy is $(1+\mathrm{TV}(P^m, Q^m))/2$: choosing Q on event D yields accuracy $1/2+(Q^m(D)-P^m(D))/2$, maximized by the likelihood-ratio event. Writing $\delta = \mathrm{TV}(P, Q)$ and $\rho = \sum_c \sqrt{P(c)Q(c)}$, we use

$$\max\{\delta, 1 - \rho^m\} \leq \mathrm{TV}(P^m, Q^m) \leq \min\{1 - (1 - \delta)^m, \sqrt{1 - \rho^{2m}}\}.$$

The first upper bound follows by independent maximal couplings; the affinity factorizes, giving the Hellinger bounds by Cauchy–Schwarz. The lower bound δ follows by projection to one coordinate. These are exact inequalities for known laws, not certificates obtained by inserting uncertain estimates.

Sampling and uncertainty. The prespecified states use $K = 4, 11, 51, 101, 201, 501$, with all-free, singleton, two/three-color, contiguous, alternating and scattered available sets. Legal local-list witnesses extend high-color fixed neighbors with private pending leaves. Additional frozen autonomous states retain their generator and graph provenance and are secondary analyses. Every policy is queried on the same frozen state. We report deterministic midpoint and boundary probes separately from IID Monte Carlo; quadrature does not receive binomial intervals.

Each network/state has 32,768 distribution-estimation draws and 8,192 independent shared-r fidelity/test draws. For six prespecified contexts and all thirty primary networks, candidate-only likelihood-ratio tests use independent fresh candidate samples at $m = 1, 5, 20, 100, 500, 2000,$ 256 test trials per source, and 1,024 independent teacher calibration trials. Test trials use nested prefixes over m, so points on a curve are paired. The plug-in alternative uses pseudocount 1/2 per palette color: an unobserved category is not declared impossible. Teacher support is known exactly. Equal-prior accuracy uses threshold zero with randomized ties; level-0.05 power uses a separately calibrated conservative threshold. Neither is an exact Bayes classifier for an unknown network law.

For N IID draws on K categories, the subset characterization of TV and Hoeffding's inequality give $\Pr(\mathrm{TV}(\widehat{q}, q) \geq \epsilon) \leq 2^K \exp(-2N\epsilon^2)$. This provides a conservative pointwise uncertainty radius for distances from the known teacher law. Approximate equivalence requires an upper confidence bound at most the prespecified TV tolerance 0.05; failure to reject is not equivalence. We also report JS, Hellinger, entropy, inverse collision probability, largest mass, forbidden mass and shared-r rank error, before aggregation by network/state. Float64 teacher inputs and clipped float32 model inputs are compared explicitly at boundaries. The primary shared-r fidelity supplies the same clipped float32 draw to both candidate functions; a separate column uses the unconverted float64 teacher draw. This prevents input-precision differences from being described as a learned candidate mechanism.

For all-free $K = 201$, normalized had mean estimated TV 0.046, shared-r fidelity 0.053, and 194.6 effective colors (inverse collision probability). For all-free $K = 201$, deepsets had mean estimated TV 0.164, shared-r fidelity 0.021, and 63.2 effective colors (inverse collision probability). For all-free $K = 201$, transformer had mean estimated TV 0.291, shared-r fidelity 0.020, and 23.7 effective colors (inverse collision probability). For all-free $K = 201$, e1-original had mean estimated TV 0.064, shared-r fidelity 0.038, and 173.2 effective colors (inverse collision probability). The original E1 networks are a separate historical cohort, not the new NORMALIZED replicas. 13/465 model–state estimates for deepsets contain forbidden proposals; the largest observed mass is 0.152 in k501-SINGLE-first. These are proposal errors; the MAX COMMIT certificate rejects the corresponding fixed-color conflicts. 13/465 model–state estimates for transformer contain forbidden proposals; the largest observed mass is 0.559 in k501-SINGLE-last. These are proposal errors; the MAX COMMIT certificate rejects the corresponding fixed-color conflicts. Estimation uncertainty and per-state sample-based classification curves are reported separately; low power is not a proof of equality.

Proposition 6 (Singleton teacher product laws)**.** *If $A_v = \{a\}$ and $\epsilon = 1 - q_M(a \mid S)$, then $\mathrm{TV}(q_L^m, q_M^m) = 1 - (1-\epsilon)^m$. Declaring the model exactly when any observed candidate differs from a has FPR zero, power $1 - (1-\epsilon)^m$, and optimal equal-prior balanced accuracy $1 - \frac{1}{2}(1-\epsilon)^m$.*

Proof. The null product law is a point mass at $(a, \ldots, a)$, whose alternative mass is $(1-\epsilon)^m$. TV

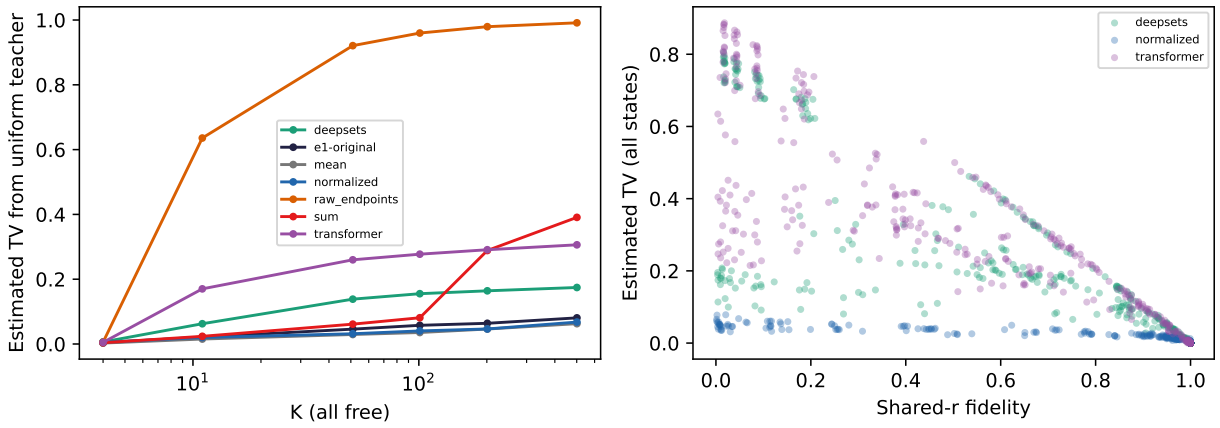


Figure 7: Conditional candidate-law distance versus palette size, and shared-r fidelity versus law distance. Each point derives from a fixed state; quadrature and Monte Carlo are stored separately.

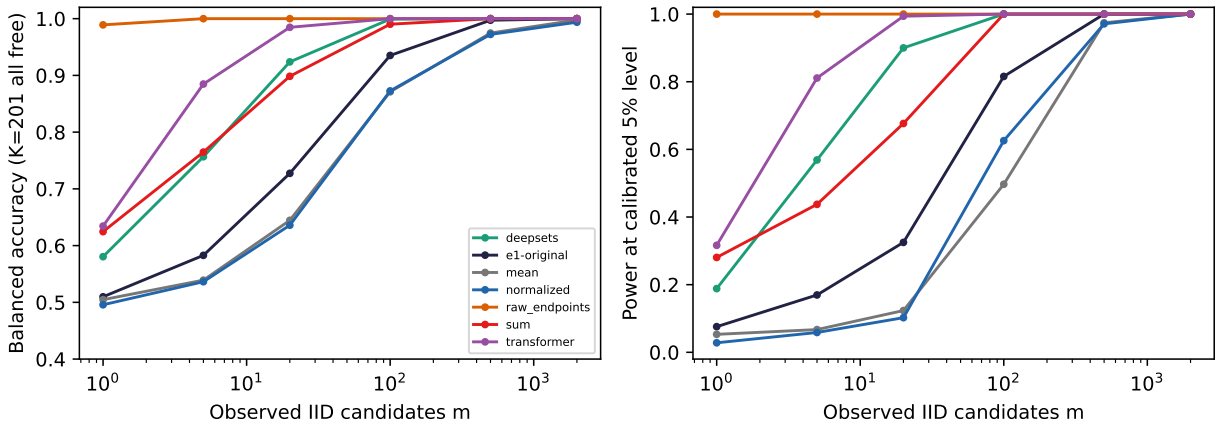


Figure 8: Independent candidate-only test performance for all-free $K = 201$. The curves summarize fixed checkpoints, not learned-law Bayes certificates.

from a point mass is one minus that mass. All other sequences have null probability zero; at the all-a sequence the null likelihood is at least the alternative likelihood. The stated rule is therefore optimal, including the identical-law tie case. □

An exploratory analytic readout of singleton states illustrates the effect. Transformer seed 41002 proposed a forbidden color with estimated probability 0.460 when only the last color at $K = 201$ was free. For five IID candidates, the support test has estimated power 0.954, with simultaneous lower/upper bounds [0.949,0.959] and exact null FPR zero. The formula is exact in the unknown true mass; these numerical values are estimates. Bonferroni Clopper–Pearson intervals cover all 1,015 recorded singleton model–state laws; this readout does not add empirical five-sample test trials.

A prospectively committed supplement applies the same design to the five original E1 checkpoints, producing 35 networks in the conditional-law diagnostics.

Distribution and progress. For exact COMMIT and conditionally private independent actions, formula (1) in the accompanying theory document applies even to heterogeneous lists. A useful TV perturbation consequence is the following: if q_v is supported on L_v, $\epsilon_v = \mathrm{TV}(q_v, \mathrm{Unif}(A_v))$ and $p_u \le p_{\max}$,

$$P_v \ge p_v \max\{0, 1 - p_{\max} t_v / k_v - \epsilon_v\}.$$

Indeed the probability of surviving neighbors' offers is a function of the candidate in $[0, 1]$; changing its expectation from uniform costs at most TV. Thus at $p = 1/2$ sufficiently small nonuniformity retains a positive constant bound, even if statistically detectable. Conversely, deterministic proposals on an all-free clique have the same one-node TV and entropy whether every node proposes a common color or different mutually compatible colors; their confirmation probabilities can be 2^{-K} or $1/2$. These marginal distances do not identify which vertices' proposals overlap. The distinct deterministic candidates here form a mathematical per-vertex distribution family; this is not claimed implementable by initially identical anonymous deterministic nodes. Marginal distance alone therefore cannot universally rank speed. The controlled experiments substitute uniform/learned/mixture candidate laws at the same state and exact COMMIT; autonomous round-count correlations remain descriptive. normalized, learned-estimated: mean exact-COMMIT probability 0.2994. normalized, half-uniform-mixture: mean exact-COMMIT probability 0.3028. deepsets, learned-estimated: mean exact-COMMIT probability 0.2709. deepsets, half-uniform-mixture: mean exact-COMMIT probability 0.2893. transformer, learned-estimated: mean exact-COMMIT probability 0.2380. transformer, half-uniform-mixture: mean exact-COMMIT probability 0.2758. These are controlled plug-in predictions from estimated laws, with fixed participation $1/2$; the heterogeneous-state experiment independently checks one-round confirmations. The largest original heterogeneous-state residual triggered a post-hoc independent replication, which predicted 165.48 versus 164.83 confirmations (observation-only standard error 0.29, 1,024 trials). Prediction uncertainty is additional; the original 256-trial results remain unchanged.

12 Discussion, Limitations and Future Work

Success on finite graphs is evidence about a specified distribution, rather than universal correctness. Five training networks permit only limited inference over initialization; small graph strata have wide confidence intervals. Shared graphs and repeated clique tapes must retain their dependence in analysis. The generic baselines retain UCGNN's graph-interface and safety pathway, so they do not test every possible end-to-end attention architecture. A fixed parameter count still permits costs that grow with N, $|E|$ and K.

An exploratory alternative teacher starts at $\lfloor K r_c \rfloor$ and scans cyclically to the first available color, keeping participation and rejection unchanged. Its candidate mass at c is the cyclic gap preceding c divided by K, usually different from uniform over available colors. A fresh seed-41005 pilot reached 0.9903 local candidate validation fidelity and completed 304/304 graph evaluations without unsafe COMMIT. On its own graph trajectories, cyclic-teacher candidate fidelity declined from 0.9911 at degree 3 to 0.03975 at degree 200, while coloring remained successful. The new candidate law is thus learned in degree and transfers as a useful coloring policy, not as an exact high-degree implementation of that teacher. This single-seed study is exploratory. It uses newly generated cyclic teacher trajectories and differs in example/update counts from the main uniform-teacher comparisons. Separate CSV columns distinguish fidelity to the cyclic and uniform teachers; they must not be pooled. Teacher-free RL and supervised-to-RL fine-tuning ask separate questions; the accompanying design specifies rewards, private stochastic actions and a PPO-based comparison

[8], with no large RL run here. Asynchronous execution, compressed communication, certified candidate anti-concentration and larger independent cohorts remain future work.

Our evidence refutes necessity of the original prefixes for the tested architecture family, and supports normalization and multiplicity control under one fixed degree-three training procedure. It does not establish necessity of MAX across all possible learned distributed algorithms, or that attention has equal population performance. Isolating the first and second communication reductions, studying more training procedures and certifying diffuse candidate mass are useful next experiments. Per-round records retain proposal support size, maximum mass and pair collisions; full candidate histograms are retained for controlled local grids, rather than every graph round. The code and fixed evaluation cohort remain available for replay and finer distributional diagnosis.

A Training runs

B Additional Local-Palette and Candidate-Law Diagnostics

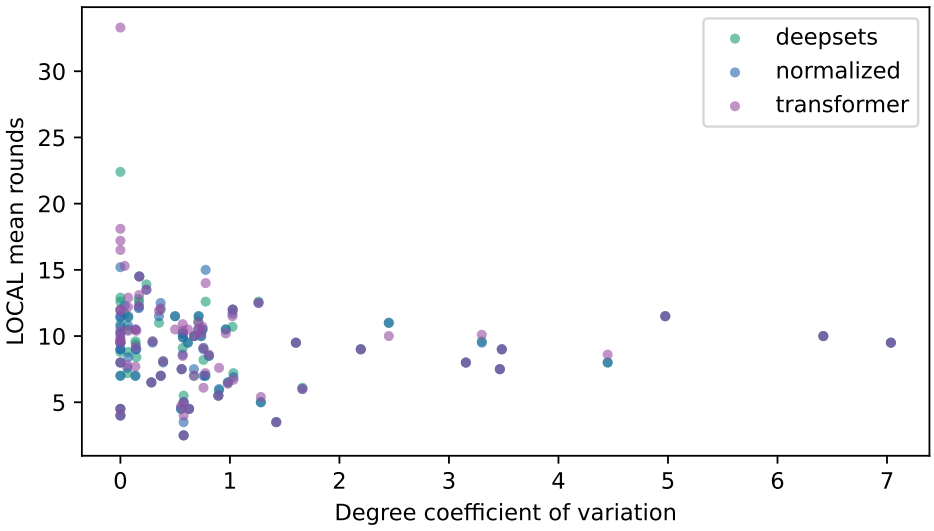


Figure 9: LOCAL mean rounds by degree heterogeneity. Points share graph identities across architectures. This association alone is not a causal effect.

References

[1] Pim de Haan, Dinesh Jayaraman, and Sergey Levine. Causal confusion in imitation learning. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://proceedings.neurips.cc/paper_files/paper/2019/hash/947018640bf36a2bb609d3557a285329-Abstract.html.

[2] Amirhossein Kazemnejad, Inkit Padhi, Karthikeyan Natesan Ramamurthy, Payel Das, and Siva Reddy. The impact of positional encoding on length generalization in transformers, 2023. URL https://arxiv.org/abs/2305.19466.

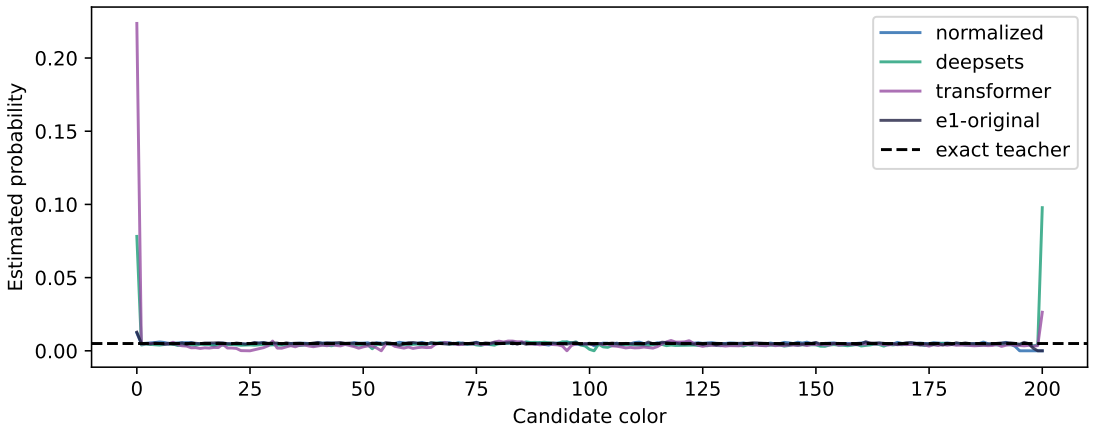


Figure 10: Estimated candidate probabilities in the same all-free $K = 201$ state, seed 41000 for each new architecture. The teacher line is analytically uniform. All five replicas and the separately labeled original E1 cohort are in CSV.

[3] Henrique Lemos, Marcelo Prates, Pedro Avelar, and Luis Lamb. Graph colouring meets deep learning: Effective graph neural network models for combinatorial problems, 2019. URL https://arxiv.org/abs/1903.04598.

[4] Andreas Loukas. What graph neural networks cannot learn: Depth vs width. In *International Conference on Learning Representations*, 2020. URL https://openreview.net/pdf?id=B1l2bp4YwS.

[5] Joël Mathys, Florian Grötschla, Kalyan Varma Nadimpalli, and Roger Wattenhofer. Flood and echo net: Algorithmically aligned GNNs that generalize, 2023. URL https://arxiv.org/abs/2310.06970.

[6] Stephane Ross, Geoffrey Gordon, and Drew Bagnell. A reduction of imitation learning and structured prediction to no-regret online learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics*, volume 15 of *Proceedings of Machine Learning Research*, pages 627–635, 2011. URL https://proceedings.mlr.press/v15/ross11a.html.

[7] Ryoma Sato, Makoto Yamada, and Hisashi Kashima. Approximation ratios of graph neural networks for combinatorial problems. In *Advances in Neural Information Processing Systems*, volume 32, 2019. URL https://arxiv.org/abs/1905.10261.

[8] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms, 2017. URL https://arxiv.org/abs/1707.06347.

[9] Olga Solodova, Nick Richardson, Deniz Oktay, and Ryan P. Adams. Graph neural networks gone hogwild. In *International Conference on Learning Representations*, 2025. URL https://arxiv.org/abs/2407.00494.

[10] Jukka Suomela. Survey of local algorithms. *ACM Computing Surveys*, 45(2):24:1–24:40, 2013. doi: 10.1145/2431211.2431223. URL https://www.cs.helsinki.fi/local-survey/.

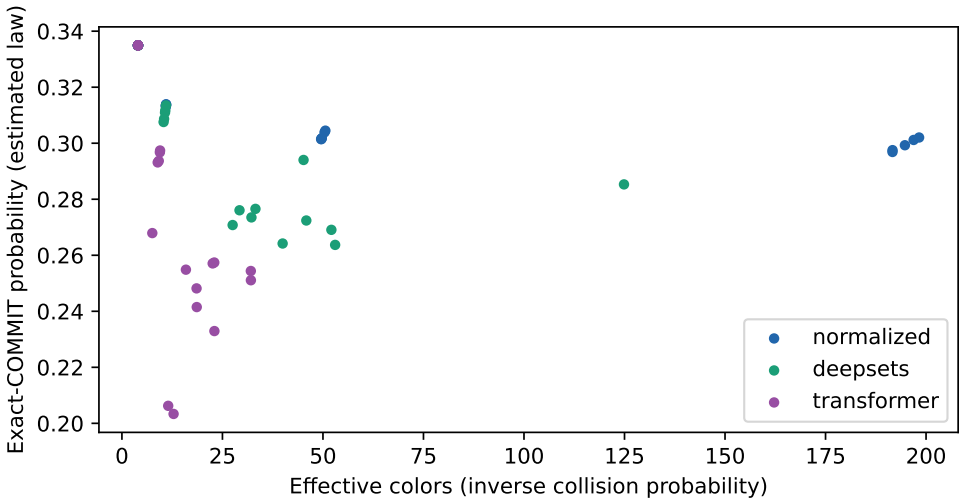


Figure 11: Candidate diversity and predicted exact-COMMIT probability on shared all-free clique states. Predictions use estimated laws and fixed participation probability 1/2, not a universal guarantee from the diversity metric.

[11] Ashish Vaswani, Noam Shazeer, Niki Parmar, Jakob Uszkoreit, Llion Jones, Aidan N. Gomez, Lukasz Kaiser, and Illia Polosukhin. Attention is all you need, 2017. URL https://arxiv.org/abs/1706.03762.

[12] Petar Veličković and Charles Blundell. Neural algorithmic reasoning, 2021. URL https://arxiv.org/abs/2105.02761.

[13] Petar Veličković, Adrià Puigdomènech Badia, David Budden, Razvan Pascanu, Andrea Banino, Misha Dashevskiy, Raia Hadsell, and Charles Blundell. The CLRS algorithmic reasoning benchmark. In *Proceedings of the 39th International Conference on Machine Learning*, volume 162 of *Proceedings of Machine Learning Research*, pages 22084–22102, 2022. URL https://proceedings.mlr.press/v162/velickovic22a.html.

[14] Keyulu Xu, Mozhi Zhang, Jingling Li, Simon S. Du, Ken-ichi Kawarabayashi, and Stefanie Jegelka. How neural networks extrapolate: From feedforward to graph neural networks. In *International Conference on Learning Representations*, 2021. URL https://openreview.net/forum?id=UH-cmocLJC.

[15] Manzil Zaheer, Satwik Kottur, Siamak Ravanbakhsh, Barnabas Poczos, Ruslan Salakhutdinov, and Alexander J. Smola. Deep sets. In *Advances in Neural Information Processing Systems*, volume 30, 2017. URL https://proceedings.neurips.cc/paper_files/paper/2017/hash/f22e4747da1aa27e363d86d40ff442fe-Abstract.html.

[16] Zhongyuan Zhao, Gunjan Verma, Chirag Rao, Ananthram Swami, and Santiago Segarra. Distributed scheduling using graph neural networks, 2020. URL https://arxiv.org/abs/2011.09430.

[17] Hattie Zhou, Arwen Bradley, Etai Littwin, Noam Razin, Omid Saremi, Josh Susskind, Samy Bengio, and Preetum Nakkiran. What algorithms can transformers learn? a study in length generalization, 2023. URL https://arxiv.org/abs/2310.16028.

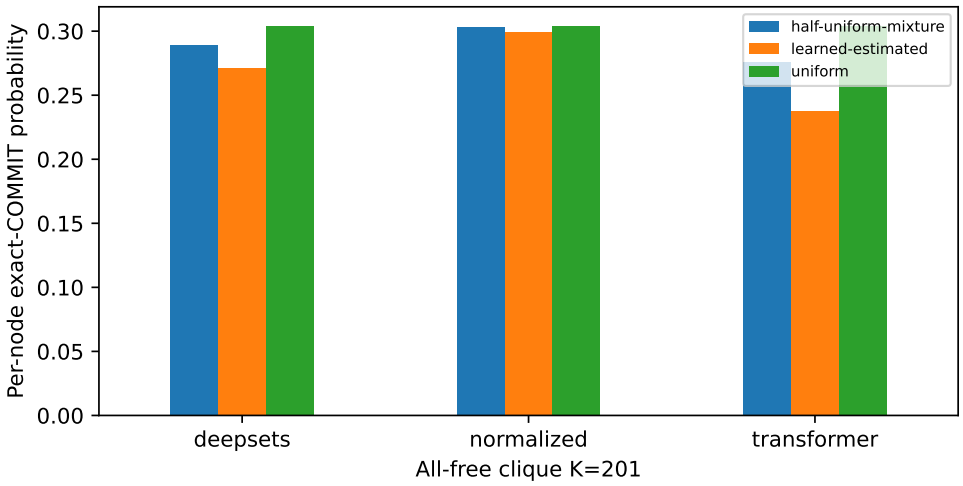


Figure 12: Controlled candidate-law substitution in the all-free $K = 201$ clique: estimated learned law, uniform law, and their 50:50 mixture. COMMIT is exact and participation is fixed at 1/2 for this intervention.

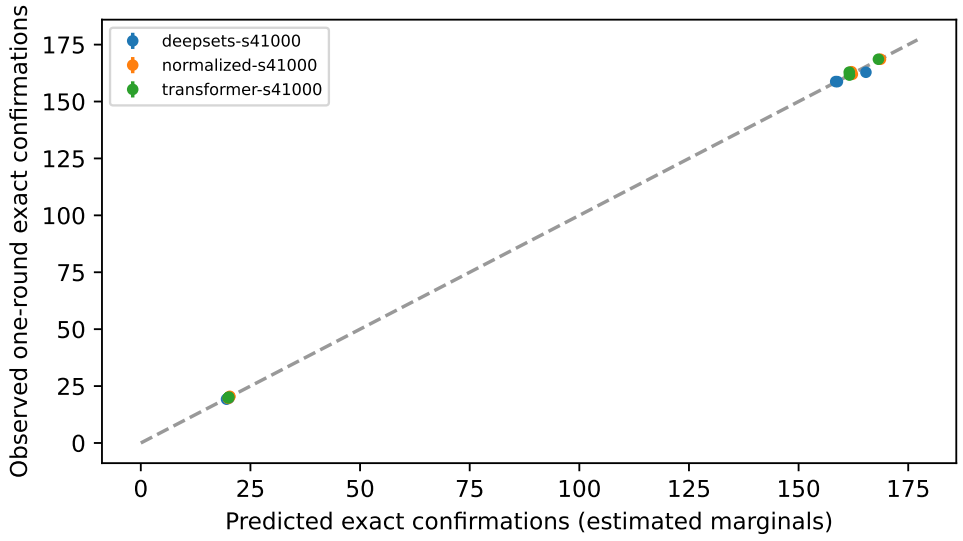


Figure 13: Six frozen heterogeneous LOCAL states, three new seed-41000 policies. Predictions from independently estimated candidate/participation marginals are compared with 256 fresh one-round exact-COMMIT observations per context. Error bars show 1.96 Monte Carlo standard errors for the observations; they do not include uncertainty in the predicted marginals.

Model	Seed	Epoch	Params	Updates	Candidate	COMMIT
cyclic_teacher	41005	120	16035	17760	0.9903	1.0000
deepsets	41000	120	17907	17880	0.9948	1.0000
deepsets	41001	120	17907	17640	0.9965	1.0000
deepsets	41002	120	17907	17760	0.9970	1.0000
deepsets	41003	120	17907	17760	0.9881	1.0000
deepsets	41004	120	17907	17760	0.9935	1.0000
mean	41000	120	16035	17880	0.9983	1.0000
mean	41001	120	16035	17640	0.9981	1.0000
mean	41002	120	16035	17760	0.9995	1.0000
mean	41003	120	16035	17760	0.9980	1.0000
mean	41004	120	16035	17760	0.9994	1.0000
normalized	41000	120	16035	17880	0.9971	1.0000
normalized	41001	120	16035	17640	0.9985	1.0000
normalized	41002	120	16035	17760	0.9986	1.0000
normalized	41003	120	16035	17760	0.9985	1.0000
normalized	41004	120	16035	17760	0.9993	1.0000
own_only	41000	120	16035	17880	0.9971	1.0000
raw_endpoints	41000	120	16035	17880	0.9850	1.0000
raw_endpoints	41001	120	16035	17640	0.9897	1.0000
raw_endpoints	41002	120	16035	17760	0.9891	1.0000
raw_endpoints	41003	120	16035	17760	0.9882	1.0000
raw_endpoints	41004	120	16035	17760	0.9886	1.0000
sum	41000	120	16035	17880	0.9983	1.0000
sum	41001	120	16035	17640	0.9980	1.0000
sum	41002	120	16035	17760	0.9991	1.0000
sum	41003	120	16035	17760	0.9982	1.0000
sum	41004	120	16035	17760	0.9993	1.0000
transformer	41000	120	16995	17880	0.9957	1.0000
transformer	41001	120	16995	17640	0.9925	1.0000
transformer	41002	120	16995	17760	0.9962	1.0000
transformer	41003	120	16995	17760	0.9932	1.0000
transformer	41004	120	16995	17760	0.9969	1.0000

Table 7: Training progress and in-degree validation, with all seeds shown. Partial epochs are explicitly listed.